

Qatar-2b

R-1913

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140401 · created 2026-08-05T23:49:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456747.44524 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.227 +/- 0.067
Transit depth (Rp/Rs)^2	5.16 +/- 3.03 [%]
Orbital Inclination (inc)	89.52 +/- 1.73
Airmass coefficient 1 (a1)	0.74 +/- 0.66
Airmass coefficient 2 (a2)	0.2 +/- 1.03
Scatter in the residuals of the lightcurve fit is	108.7 %
Best Comparison Star	#3 - [572, 297]
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.0251 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.227 $\pm$ 0.067 vs 0.16327 (deviation 0.95 σ)
Duration fit vs published	0.0251 d vs 0.0767 d (deviation 21.49 σ)
Mid-time O-C	-1925.1 min
Depth SNR	3.4
Residual scatter	108.7 %

Light curve

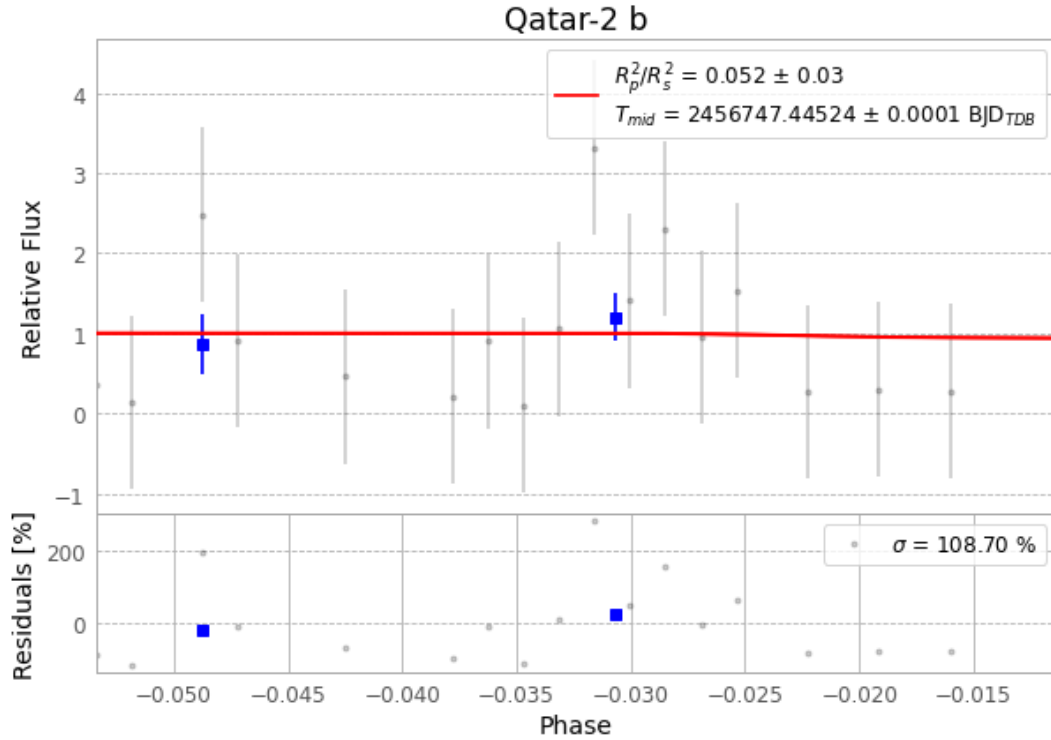


Figure 1 — FinalLightCurve_Qatar-2 b_2014-03-31.png (SHA-256 8bb8fd4319651fb2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [158, 448], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef0b5231bbed8653...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), Qatar-2140401045412.FITS → Qatar-2140401084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140401.log (67e0b43d2872...), FinalLightCurve_Qatar-2 b_2014-03-31.png (8bb8fd431965...), FinalLightCurve_Qatar-2 b_2014-03-31.csv (0bc70d5e066a...)

Compute resources

Wall time 147.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:49Z.